

Photovoltaic H6 -Type Transformerless Inverter Topology

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Abstract— The concept of transformerless inverters has become a future trend for single phase photovoltaic grid - tied systems. The major factor to be considered while employing transformerless inverters in such applications is the leakage currents. The lack of transformers in inverters, implemented between PV panel and grid results in a galvanic connection and thus there arises problems due to leakage currents. This paper deals with an H6 transformerless full- bridge inverter topology with low leakage currents that can be used in PV grid tied applications. This H6 inverter topology is taken as an example for detail analysis with operation modes and modulation strategy. A closed loop has been developed for maintaining the voltage constant at the grid side irrespective of changes in the input voltage. Aforementioned transformerless topology is simulated for evaluating the performances in terms of power efficiency and leakage currents characteristics and a prototype for the H6 inverter topology is also built.

Key Words - Common-mode voltage, grid-tied inverter, leakage current, photovoltaic (PV) generation system, transformerless inverter.

I. INTRODUCTION

Inverters with transformers of conventional type, connected in PV grid-tied generation systems have now being replaced by transformerless inverters due to various reasons such as reduction in size, weight and cost, improvement in efficiency etc. Transformerless inverters cause a number of technical challenges in grid-connected PV systems, among which flow of leakage currents is a major problem. This leakage currents that flows between the parasitic capacitance of PV array and the grid has to be eliminated, which otherwise leads to serious safety problems. A photo - voltaic system is a power generation system designed to harness and supply usable solar power by means of photo-voltaic technology. It consists of an arrangement of several components, including solar panels to absorb and directly convert solar energy into electrical energy, a solar inverter to convert the electrical current from DC to AC, other electrical accessories and power conditioning units to organize a working system. An electrical grid is an interconnected network for delivering electricity from suppliers to consumers. It consists of generating stations that produce electrical power, high-voltage transmission lines that carry power from distant sources to demand centers, and distribution lines that connect to individual customers.

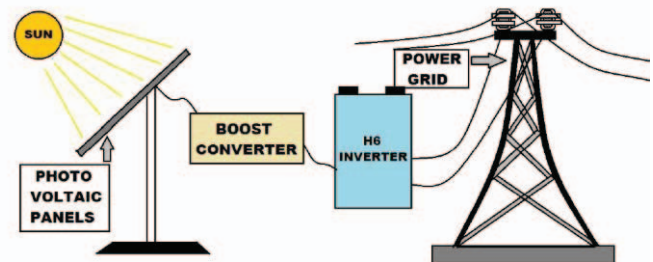


Fig. 1. Schematic representation

The Schematic representation for the proposed system is shown in fig.1.

As a result of increasing population growth and industrial development energy shortage is a major crisis that exists today. If we can utilize the renewable energy resources which do not get depleted with time and which are very environment friendly with no pollution, this energy crisis can be reduced to a considerable extent. Solar energy is one of the most important renewable energy resources which is abundantly available all over the world. With reduction in the system cost the PV technology has the potential to become one of the main renewable energy sources for the future electricity supply. Photovoltaic (PV) power supplied to the utility grid is gaining more and more visibility, while the world's power demand is increasing. Solid-state inverter seems to be the enabling technology for putting PV systems into the grid. Thus an inverter which can effectively link the solar generator and the public power grid is a topic that should be dealt with.

II. PROPOSED TOPOLOGY

The circuit diagram for the proposed H6 inverter topology is shown in fig.2. It consists of six MOSFET switches S_1, S_2, S_3, S_4, S_5 and S_6 . The modulation technique used is the unipolar sinusoidal PWM. The photovoltaic panel is represented by V_{pv} . An LCL filter is used for filtering purpose. V_{grid} represents the electrical grid. C_{in} represents the input dc link capacitance.

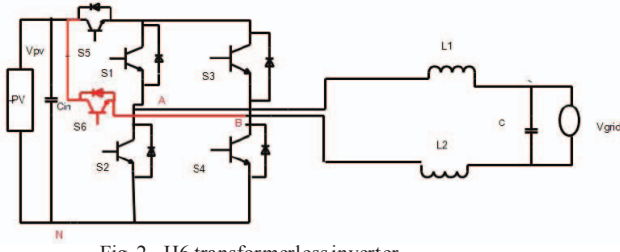


Fig. 2. H6 transformerless inverter

The proposed inverter topology, which can minimize the leakage currents by maintaining the common mode voltages constant can be interconnected in PV grid tied systems. In conventional H bridge inverter topologies, there is no need of maintaining the common mode voltages constant as there are no major issues of leakage currents, since they utilize inverters with transformers. The avoidance of transformers causes galvanic connection between PV and grid, thereby resulting in leakage currents. In such a situation the leakage currents can be reduced by maintaining common mode voltages constant.

A. Basic operating principle

PV panel tracks solar rays using solar cells. The dc voltage obtained is boosted to a higher level using a boost converter. Then the boosted dc voltage is given to the proposed transformerless full bridge H6 inverter which converts the DC voltage to AC. Two filter inductors L_1 and L_2 and a filter capacitor C is used for filtration. The AC power is then fed to the electrical grid.

B. Assumptions

- Filter inductors used L_1 and L_2 are assumed to be of same value.
- Common mode voltage, V_{CM} , that should be maintained constant in order to reduce leakage currents is expressed as $V_{CM} = (V_{AN} + V_{BN})/2$
- The switches and diodes are ideal and the dead time between the switches are neglected.
- Inductors are ideal without any internal resistance

III. MODES OF OPERATION

There are four modes of operation for the proposed H6 inverter. The four modes include two active modes and two freewheeling modes. The four modes of operations are shown in fig.3,4,5 and 6 respectively.

Mode 1 - Active mode

During active mode of positive half period switches S_1 , S_4 and S_5 will conduct and switches S_3 and S_6 remains in the off position. The direction of current flow is indicated by the dark black lines. The common mode voltage V_{CM} is given by $V_{CM} = (V_{AN} + V_{BN})/2 = 0.5V_{PV}$ $V_{AN} = V_{PV}$ $V_{BN} = 0$; thus $V_{AB} = V_{PV}$

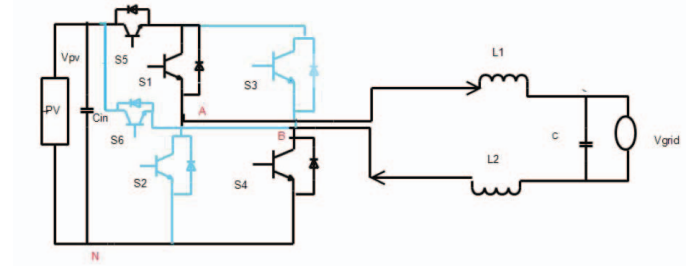


Fig. 3. mode 1

Mode 2 – Freewheeling mode

During freewheeling mode of positive half period, current freewheels through switch S_1 and the antiparallel diode S_3 . All the other switches remains in off position. The common mode voltage during this period is given by $V_{CM} = V_{AN} + V_{BN}/2 = 0.5V_{PV}$. $V_{AN} = V_{BN} = 0.5U_{PV}$ Thus, $V_{AB} = 0$

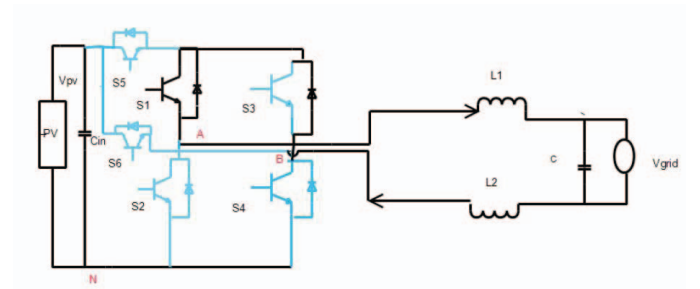


Fig. 4. mode 2

Mode 3 - Active mode

During active mode of negative half period switches S_2 , S_3 and S_6 will conduct and switches S_1 and S_4 remains in the off position. In this mode of operation although three switches are turned on, current flows through only S_2 and S_6 and hence the conduction losses can be reduced. The direction of current flow is indicated by the dark black lines. The common mode voltage V_{CM} is given by $V_{CM} = V_{AN} + V_{BN}/2 = 0.5V_{PV}$. $V_{AN} = 0$, $V_{BN} = U_{PV}$ thus, $V_{AB} = -U_{PV}$

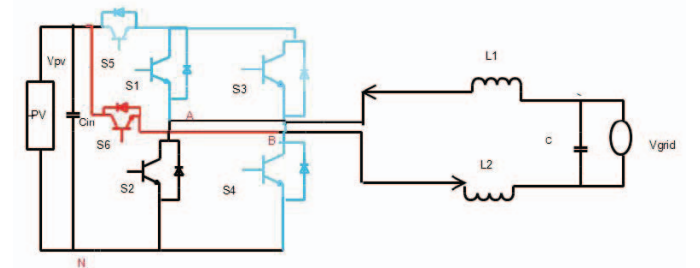


Fig. 5. mode 3

Mode 4 -Freewheeling mode

During freewheeling mode of negative half period, current freewheels through switch S_3 and the antiparallel diode S_1 . All the other switches remains in off position. The common mode voltage during this period is given by $V_{CM} = V_{AN} + V_{BN}/2 = 0.5V_{PV}$. $V_{AN} = V_{BN} = 0.5U_{PV}$ thus, $V_{AB} = 0$

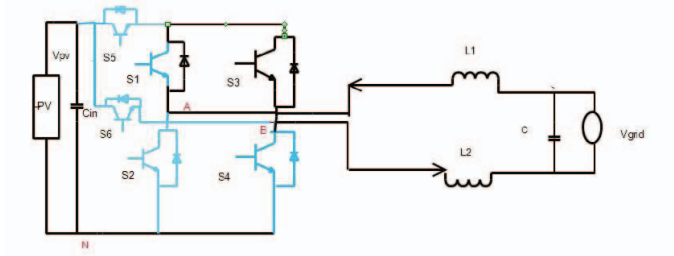


Fig. 6. mode 4

Gate signals for the inverter are shown in fig.7.

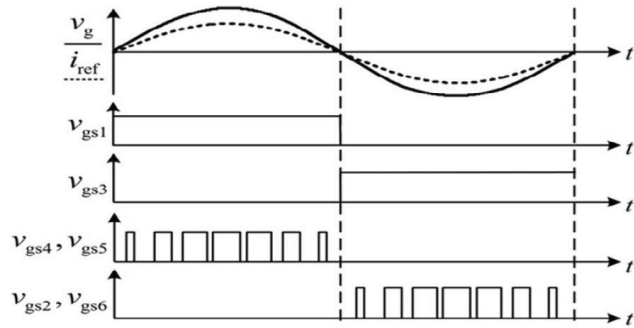


Fig. 7. Gate drive signals

IV. SIMULATION ANALYSIS

A. Circuit specification

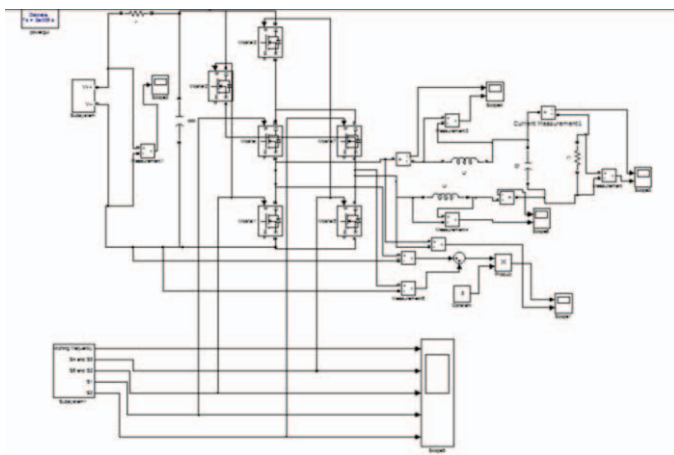


Fig. 8.a. Simulation Diagram with resistive load

By using MATLAB/SIMULINK, proposed system is simulated. Figures 8.a and 8.b shows the simulation diagrams for closed loop and open loop respectively. In open loop simulation, grid is connected at the load side. The results of open loop are similar to that of closed loop simulations.

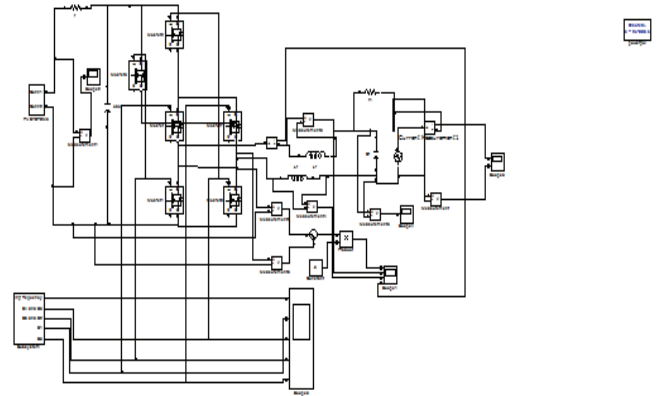


Fig. 8.b. Simulation Diagram with grid

The closed loop developed for maintaining the voltage constant is shown in fig.9.

B. Discussions on simulation results

In closed loop simulation the grid is replaced by a resistive load since in hardware a resistive load is used. The closed loop controls and maintains the output voltage at 230 V irrespective of the changes in the input side. In closed loop simulation a 72 V dc is obtained from the PV panel. Then the voltage is boosted to 230 V DC using a boost converter and converted to 230 V AC by the H6 inverter and it is connected to a resistive load. An output voltage of 230 V and current of about 0.25 A is obtained.

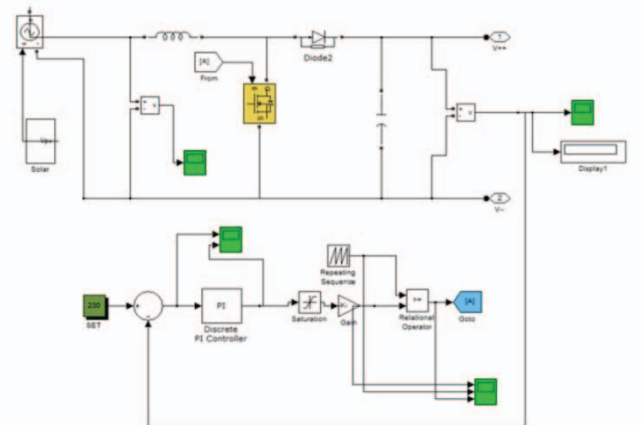


Fig. 9. Closed loop developed for voltage control

The simulation results obtained are shown in fig 10,11,12 respectively.

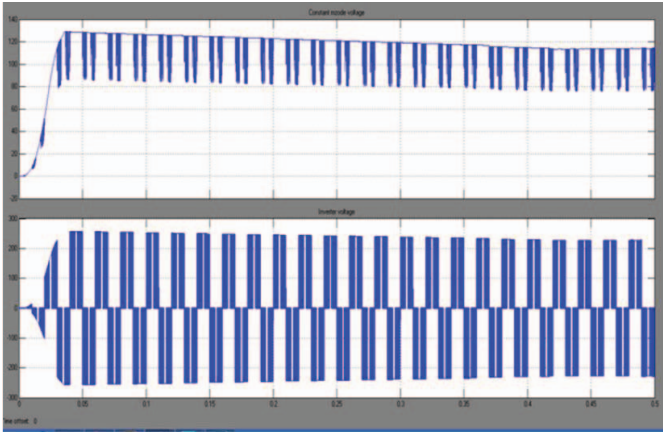


Fig. 10. Waveforms for common mode voltage and inverter output voltage

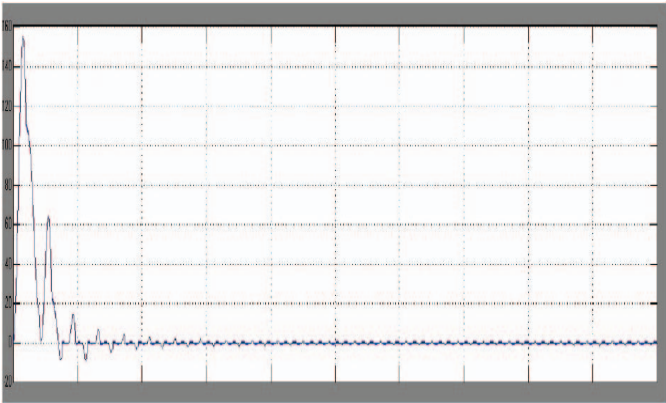


Fig. 11. Waveforms for I_1



Fig. 12. Waveform for grid voltage

V. HARDWARE IMPLEMENTATION

Hardware consists of mainly three parts:

- _Power circuit
- _Driver circuit
- _Control

circuit

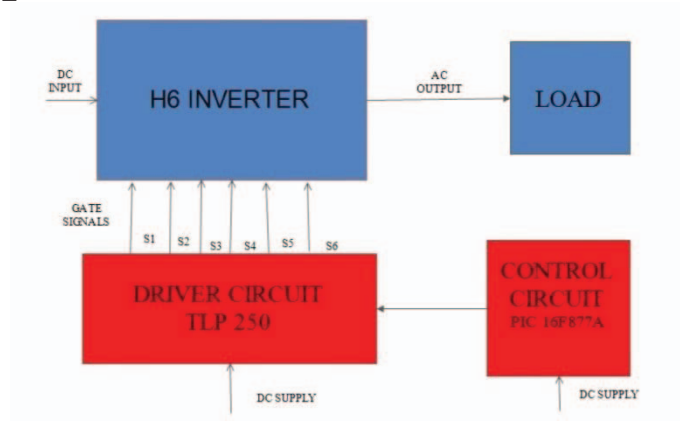


Fig. 13. Block diagram of representation

The block diagram for the hardware circuit is shown in fig.13.

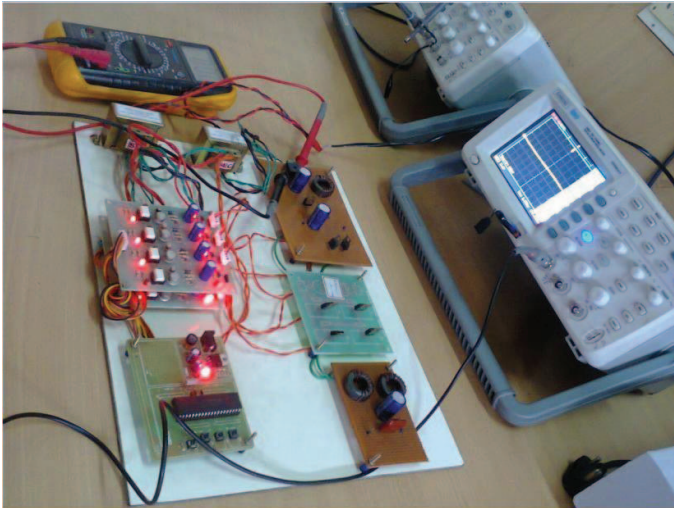


Fig. 14. Experimental setup

The Experimental setup for the prototype model of H6 transformerless inverter model is shown in fig.14.

A. Analysis of output

Two 230/12V transformers are connected in series.230V supply is step down to 12V DC and rectified using a bridge rectifier .Thus a 24V dc is given as the input. This 24 V is booted to a value of about 50V using a boost converter and given to the H6 inverter which converts the DC to AC. Even if the output voltage obtained is AC , it is not completely sinusoidal. The hardware output obtained is a square wave AC. Voltages V_{an} and V_{bn} can be measured at terminals A and B. The voltages V_{an} and V_{bn} are obtained as square waves since no PWM is used in the hardware implementation.

The common mode voltage that has to be kept constant in order to reduce the leakage currents is given by $(V_{an} + V_{bn}) / 2$. By adding the two voltages and dividing by two a constant voltage is obtained showing that low leakage currents flows through the circuit converts the DC to AC. Voltages V_{an} and V_{bn} can be measured at terminals A and B. The common mode voltage that has to be kept constant in order to reduce the leakage currents is given by $(V_{an} + V_{bn}) / 2$. By adding the two voltages and dividing by two a constant voltage is obtained showing that low leakage currents flows through the circuit. The DSO outputs obtained are shown in fig 15,16,17,18,19 respectively.

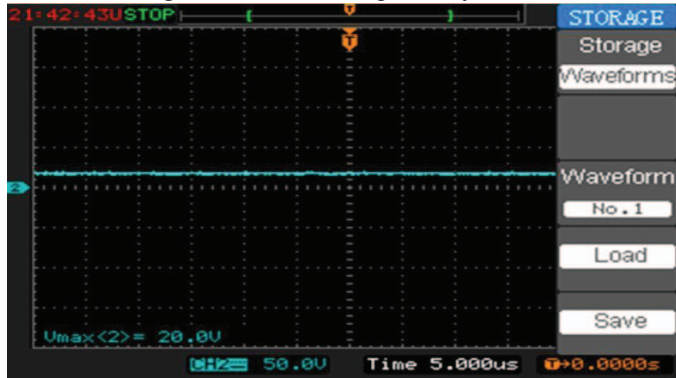


Figure 15: Input DC voltage

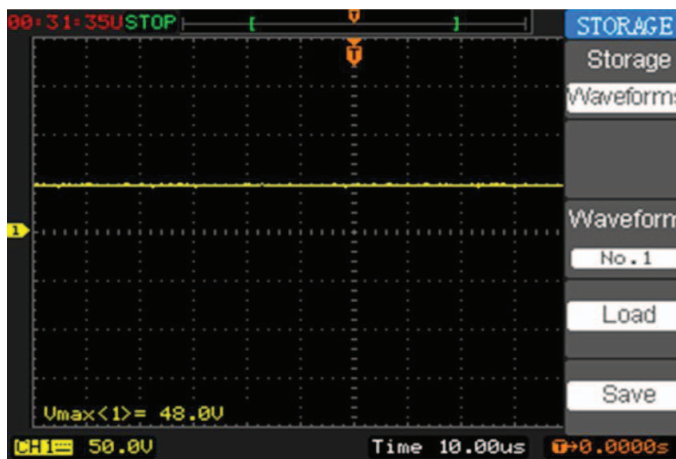


Fig.16. Boosted DC voltage

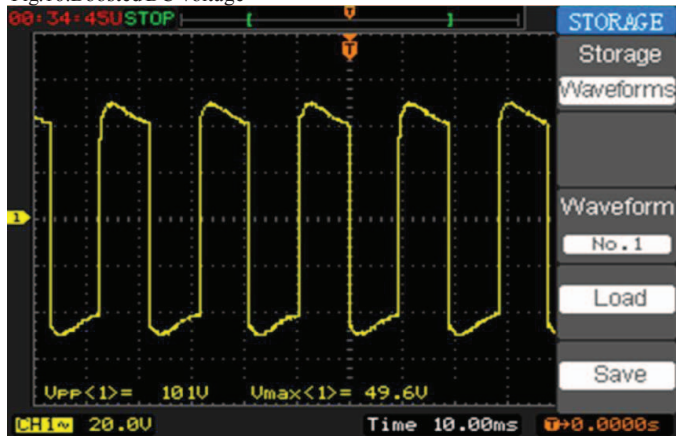


Fig.17. Output AC Voltage

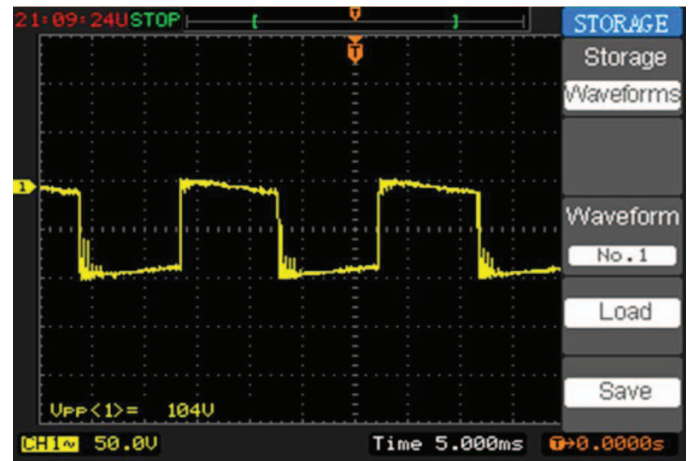


Fig.18. Van Voltage

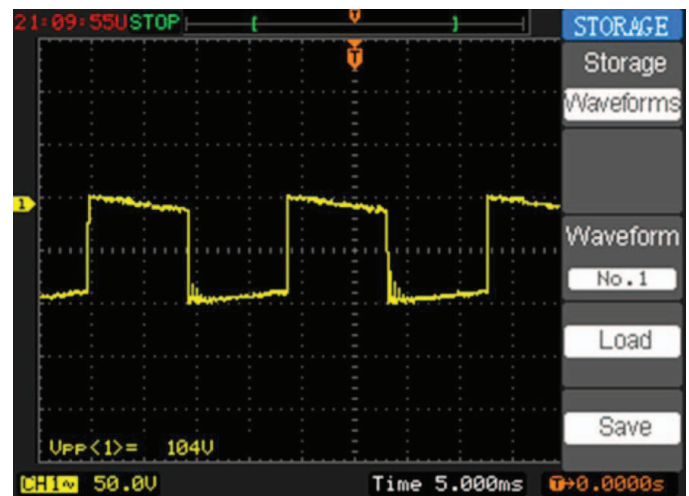


Fig.19. V bn Voltage

VI. CONCLUSION

Tapping roof top solar power is gaining importance in order to meet shortage of electric power. The solar power can be fed to the grid through an inverter. There are different topologies of inverter with and without galvanic isolation. In this work a topology, H6 topology is taken for analysis, design and simulation. The circuit is simulated and results satisfying the objectives are obtained. The leakage current is also tested through simulation. A small model of hardware set up is assembled and tested. It is found out that the common mode voltage remains constant and thus it is inferred that a reduced leakage currents can be obtained.

VII. FUTURE WORK

- Proposed system has to be analyzed with renewable source instead of DC source in the circuit.
- Synchronization of the inverter topology with grid has to be done.
- MPPT can be implemented and analyzed.
- Impact of inductive load on the system has to be dealt.

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